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Bamenda Archdiocesan Examination Board (BAEBOC)	SUBJECT CODE NUMBER 0715	PAPER NUMBER 2
MOCK G.C.E. EXAMINATION	SUBJECT TITLE AND SYLLABUS **CHEMISTRY**	
ADVANCED LEVEL	DATE MARCH 2025	
FOR EXAMINER'S USE ONLY NAME OF EXAMINER: …… SIGNATURE OF EXAMINER:…… DATE:……	FOR OFFICIAL USE ONLY	FOR EXAMINER'S USE ONLY

CANDIDATE NUMBER	
DATE/SIGNATURE	

THREE HOURS

Answer ALL the six questions in this booklet.

The mark allocation is indicated for each question. Each question carries 20 marks.

Verify that this booklet contains six questions, no questions are repeated and there are no blank pages. Inform the invigilator in case this booklet contains less than six questions, questions are repeated or there are blank pages so that the booklet should be changed.

In calculations you are advised to show all the steps in your working, giving your answer at each stage.

All necessary working must be shown. No marks will be awarded for answers without brief statements showing how the answers have been obtained.

Useful Data

Na=23, C=12,0=16, Avogadro number, 6.02×10^{23}

1 mole of gas = $24000cm^3$ ($24dm^3$)

TURNOVER

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SECTION A: PHYSICAL AND GENERAL CHEMISTRY

1. a) Define

i) A mole _____

_____

_____

ii) Avogadro number _____ (2mks)

b) 7.50g of a hydrated copper(II) sulphate ($CuSO_4.nH_2O$) on heating gave 4.8g of anhydrous copper(II) sulphate, determine the value of n (moles of water of crystallization)

_____

_____

_____

_____

_____ (2mks)

c) i) State Raoult's law

_____

_____

_____

_____

ii) A mixture of water (B.Pt 100°C) and ethanol (B.Pt78°C) deviate from Raoult's law and form an azeotropic mixture of composition 96% ethanol.

a) What is an azeotropic mixture?

_____

_____

b) Draw a well labeled B.pt/composition graph for the mixture to show the azeotrope at constant pressure.

_____

_____

_____

_____

iii) Give two ways by which the azeotropic mixture could be fairly separated

_____

_____

_____ (4mks)

d) i) What is an intermolecular force?

_____

_____

_____

ii) Use a suitable example of a substance in each case to illustrate two type of intermolecular forces

_____

_____

_____ (3mks)

e) The Bragg's equation is important in solid state chemistry

i) Write the equation _____

ii) What is a unit cell _____

_____

_____

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iii) Draw the unit cell of sodium chloride

(3mks)

f) i) Given the following standard enthalpies of combustion in $kjmol^{-1}$ $CH_2=CH_2$ = -1411.1, CH_3CH_2OH = -1367.0. Calculate the enthalpy change when 23.0g of ethanol are formed according to the equation

$C_2H_4(g)) + H_2O \longrightarrow C_2H_5OH(l)$

ii) Consider the equilibrium in a sealed tube for H_2 and nitrogen gas

$3H_{2(g)} + N_{2(g)} \longleftrightarrow 2NH_{3(g)}$ $\Delta H = -92kjmol^{-1}$

A) State the equilibrium law

B) State and explain how the value of the equilibrium constant would be affected by increase in temperature

(3mks)

g) i) Define lattice energy

ii) Draw a born haber cycle for sodium oxide (Na_2O)

(4mks)

20 MARKS

2. a) This question concern atomic structure bonding and equilibria. The first ionization energy of an element is $+1600kjmol^{-1}$

i) define first ionization energy

ii) Determine the ionization energy of one atom

(2mks)

b) A radioactive nuclei $^{131}_{15}P$ decays into a nuclide Q by emitting an alpha and beta particle . It has a half life of 3days

i) Define radioactivity

ii) Write an equation for half life

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iii) Calculate the time taken for 1.0g of P to decrease to 0.125g

(3mks)

c) Given the half cell: $Cu_{(s)}/Cu^{2+}_{(aq)}$ E=+0.34v and $Fe_{(s)}/Fe^{2+}_{(aq)}$ E=+44v

i) Give the name of an instrument that could be used to measure the emf of the cell when the two half cells are coupled

ii) What is the composition and function of a salt bridge?

iii) Draw an electrolytic cell consisting of the two half cells

(4mks)

d) A mixture of ethanoic acid (CH_3COOH) and sodium ethanoate (CH_3COONa) form a solution.

i) What name is given to this type of solution?

ii) By means of equations show how the solution above behaves on addition of a small amount of

a) $KOH_{(aq)}$

b) $HCl_{(aq)}$

(5mks)

iii) Calculate the P^H of the solution containing $0.02moldm^{-3}$ sodium ethanoate and $0.03moldm^{-3}$ ethanoic acid (ka=$1.6x10^{-5}moldm^{-3}$.

e) Mercury(ii)chloride react with oxalate ions $C_2O^{2-}_4$, in solution with a precipitate of Hg_2Cl_2 formed.

$2HgCl_2(aq) + C_2O_4^{2-}(aq) \longrightarrow 2Cl^-(aq)+2CO_2(g)+Hg_2Cl_2(s)$

The initial rate at 373k for solution of concentration expressed as $moldm^{-3}$ of $HgCl_2$ reacted as shown in table

EXP't	$[HgCl_2]moldm^{-3}$	$[C_2O_4^{2-}]moldm^{-3}$	Rate $x10^{-4}moldm^{-3}$
1	0.0836	0.202	0.52
2	0.0836	0.404	2.08
3	0.0418	0.404	1.06

I. What technique can be used to determine the rate of the reaction?

II. Determine the order of reaction with respect to

A. $HgCl_2$

B. $C_2O_4^{2-}$

III. Calculate a value for rate constant (k) and state the unit

(6mks)

20 MARKS

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SECTION B: INORGANIC CHEMISTRY

3. This question concern periodic table and periodicity of S-block elements

a) The element in period 2 (Li to Ne) of the periodic table show periodic trend

i) State and explain how atomic radius varies across the period

_____

_____

ii) Identify one common feature of the electronic configuration of the s-block element

_____

iii) Lithium is diagonally related to magnesium. Why are the two elements diagonally related?

_____

_____ (3mks)

b) i) Give the formulae of the stable chlorides of the elements Li to Ne

Li	Be	B	C	N	O	F	Ne

From the table identify the chloride with

i) Bent shape _____

ii) Linear shape _____ (4mks)

c) The compound "cisplatin" has the formula $[Pt(NH_3)_2Cl_2]$

I) Give the oxidation state of platinum in the compound

_____

II) What is the coordination number of platinum in the cisplatin?

_____

III) Give the name of the complex $Cu_2[Fe(CN)_6]$

_____ (3mks)

d) This part is on the group vii (halogens)

i) Complete the table below

Halogen	F	Cl	Br	I
Colour				

ii) Which is the most oxidizing of the elements?

_____

iii) The elements undergo displacement as well as disproportionation. Define the underlined terms.

_____

_____

_____

iv) Sodiumhyprochloride(NaOCl) is used as commercial solution. Give one use of this reagent and explain how it functions

_____

_____ (6mks)

e) This is on sulphur and nitrogen

i) Give the main source of the nitrogen in the haber process

_____

ii) State the conditions applied in the haber process to produce ammonia

_____

_____

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iii) Complete the table below by writing the formulae and names of compound (ion) in which sulphur has the oxidation states

Names of ions in oxidation state	Formula of ion compound	Name of ion compound
+4		
+2		

(4mks)

Total= 20mks

4. a) The following question concern the chemistry of group(iv) elements (carbon to lead)

i) What are the principal oxidation states of the element in their chlorides?

ii) What is the trend in the oxidation states down the group?

(2mks)

b) Define

i) Catenation

ii) Inert pair effect

iii) Using carbon and lead illustrate the definition above

(4mks)

c) State two ways in which carbon differs from the other elements in the group

(1mk)

d) i) State and explain the trend in the hydrolysis of tetrachlorides

ii) Give the formula of a group (iv)oxide which is:

acidic _____ amphoteric _____ (4mks)

e) Cobalt, Iron and manganese are d-block elements as well as transition elements

f) i) What are transition elements

ii) Iron can form a complex ion. Give the formula of one complex compound of Iron.

iii) Write the electronic configuration of Manganese(25)

iv) The ion Fe^{2+} is readily oxidized to Fe^{3+} while Mn^{2+} are not readily oxidized to Mn^{3+}. Explain.

(5mks)

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f) Explain the following observations

i) Group(ii) metals are harder than the group I

ii) Atomic radius decrease across the period in P.T

iii) Ionization energy decrease down each group

(4mks)

20 MARKS

SECTION C: ORGANIC CHEMISTRY

5. a) This question concern the following aliphatic alcohols

I) $CH_3CH_2CH_2OH$ II) $CH_3CHOHC_2H_5$ III) $CH_3-C(CH_3)(CH_3)-OH$

i) Give the names of II and III

II_____

III_____

ii) Which of the alcohols contains an asymmetric carbon atom Give its isomers.

iii) Which of them has a higher boiling Point? Explain.

iv) How will you distinguish between the alcohols above chemically?

(8mks)

b) Given the following scheme

C_2H_6O (A) $\xrightarrow{I}$ B C_2H_5Br $\xrightarrow{II}$ C C_3H_5N $\xrightarrow{III}$ D C_3H_9N $\xrightarrow{IV}$ E $C_2H_5C(=O)Cl$

i) Identify the functional group in the compounds A-E

Compound	A	B	C	D	E
function					

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ii) Give the reagents and reaction conditions for reactions I to III

	reagents	conditions
I		
II		
III		

iii) In a laboratory conversion of compound A to B, 8.79g of B were obtained from 4.81g of A calculate the percentage yield (C = 12, O = 16, H = 1)

(7mks)

c) Benzene and cyclohexene are hydrocarbonds with the same number of carbon atoms

i) How would each burn? Give the observations.

a) benzene

b) cyclohexene

ii) Write an equation for the reaction of benzene with

a) Nitrating mixture (concentrated $H_2SO_4 + HNO_3$)

b) Chloromethane

c) Outline the mechanism of the reaction between benzene and chloromethane in Aluminium chloride.

(8mks)

20 MARKS

6. a) Analysis of a certain liquid A, reveals that it contains 92.3% carbon and 7.7% hydrogen. Vaporization of 0.02g of the liquid at 99°C and 740mmHg pressure gives 80.5cm^3 of vapour Calculate

i) the empirical formula of A

ii) the molecular mass of A

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iii) the molecular formular of A

(5mks)

b) Consider the following type of organic reactions electrophilic addition nucleophilic addition and nucleophilic substitution . Given the compound $CH_3CH_2CHBrCH_3$, CH_3CHO, CH_3CHCH_2. Write suitable equations that illustrate the following reaction type by choosing from the compounds above the one that could undergo:

i) Electrophilic addition

ii) Nucleophilic addition

iii) Nucleophilic substitution

(6mks)

c) i) State Markvonikoff's rule

ii) Give the mechanism of the reaction of but-1-ene with hydrogen bromide

(3mks)

d) The molecular formula $C_5H_{10}O$, represents a number of isomeric compounds including the enol $CH_3CH{=}CHCHOHCH_3$

i) Give the systematic name of the enol:

ii) Give the different stereoisomers of the enol that are

A) Geometric isomers	B) Optical isomers

iii) Give one structural isomer of the enol.

iv) Give a chemical test to distinguish between any one pair of stereoisomers in (ii) above

(6mks)

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